

AHMAD BILAL

Data Analyst & Systems Tooling Developer

Lahore, Pakistan | kierninja@gmail.com | github.com/AhmadBilalDSA | linkedin.com/in/AHMADBILALDES

PUBLIC REPOS 47 | PRs AUTHORED 53 | MERGED 24 | OPEN 13 | FOLLOWERS 9

TECHNICAL SUMMARY

Data Analyst and systems developer specializing in high-throughput SQL, AST analysis, automated data diffing pipelines, and native desktop systems. Verified by 24 merged upstream PRs across SQLGlot, HFlow, SciTeX-io, dbt-duckdb, and tidb, plus 47 public repositories.

ENGINEERING PROJECTS

MockForge [Rust | Tauri v2 | React Flow | TypeScript | Tailwind CSS]

Native AI mock interview simulator and interactive system design workbench.
Offline heuristic feedback with zero cloud latency; SPOF / cache-pressure analyzer on an interactive canvas.
Windows SAPI speech integration delivers native on-device voice.

duck-diff [DuckDB | Python | GitHub Actions | Discord Webhooks]

Automated data diff pipeline and visual lineage engine with constant-memory reconciliation.
Headless CI/CD telemetry tracking with automated regression detection across pipelines.
Scheduled daily health dispatch to Discord; live telemetry feeds the portfolio console.

sqllean-lint PRO [Python | SQLGlot | AST parsing | CustomTkinter]

Local-first SQL static analysis and AST mutation CLI.
Multi-dialect linting, query tree inspection, and automated anti-pattern rewrites.

UPSTREAM OPEN SOURCE 53 PRs authored / 24 merged

Ibis PR #12086 (open): pl.Object conversion to dt.Unknown in PolarsType.to_ibis.
SQLGlot PR #8257: DuckDB LIST_POSITION / LIST_INDEXOF / LIST_REVERSE array AST nodes.
SymPy PR #30384: type annotations for sympy.utilities.decorator.
HFlow PR #281 (merged): consolidated curation validation rules, closing #280.
SciTeX-io PR #166 (merged): Feather I/O round-trip and invalid-extension hardening.
Also merged: pingcap/tidb #70912 (IndexMerge truncation), duckdb/dbt-duckdb #835 (Windows fix).

PROFESSIONAL EXPERIENCE

Power BI & Analytics Trainer [PNY Trainings / NAVTTC / 2026]

Instructed Star Schema dimensional modeling, advanced DAX measures, and Power Query ETL pipelines.
Delivered end-to-end business intelligence and data analytics training to NAVTTC cohorts.
Turned real-world finance datasets into hands-on Power BI labs and graded assessments.

Junior Data Scientist & Analyst [Timmy's Pizza / 2024 - 2026]

Automated operational telemetry, inventory forecasting, and data audits using Python, SQL, and Excel.
Built operational reporting pipelines, sales performance metrics, and scheduled reporting.
Designed star-schema models and DAX measures feeding executive Power BI dashboards.

EDUCATION

BS (Hons) Accounting & Finance [Hailey College of Commerce, University of the Punjab]

Graduated Nov 2024. Coursework: Financial Modeling, Quantitative Analysis, Business Statistics, Information Systems.

CERTIFICATION

IBM Data Science Professional Certificate

Python, SQL, Data Analysis, Predictive Modeling, and Machine Learning workflows.

TECHNICAL ARSENAL

Core Languages: SQL (PostgreSQL, MySQL, DuckDB, SQLite); Python (Pandas, NumPy, AST/SQLGlott); Rust (Tauri v2); TypeScript
BI & Analytics: Power BI (DAX, Star Schema, Power Query); Advanced Microsoft Excel (XLOOKUP, Pivot Tables, Macros).
Infrastructure & Automation: GitHub Actions CI/CD; Docker; Git; Linux/Bash; Headless Chromium (Playwright).